SDAM first-level analysis with identity-kernel DKGE

Identity design metric baseline on MVPA AUC maps

dkge example

2026-05-09

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1 Dataset and study design

The dataset is 32 participants (14 controls, 18 SDAM) each contributing four whole-brain MVPA AUC maps in a 2 x 2 within-subject design crossing **task** (`recog`, `nback`) with **measure** (`low_mid`, high_sem). Maps are AUC - 0.5 so they are zero-centred at chance. The shared analysis mask contains 218,414 voxels where every map is finite and nonzero.

Table 1: Four cells of the within-subject 2 x 2 design.

task	measure	cell label
recog	low_mid	recog_low_mid
recog	high_sem	recog_high_sem
nback	low_mid	nback_low_mid
nback	high_sem	nback_high_sem

2 DKGE setup

Each subject contributes a `q x P_s` matrix `B_s` whose rows are the four cell maps in canonical order. Because each cell is observed once per subject, the per-subject "design" is the `q x q` identity. The factorial structure is used for labels and contrasts, but this baseline uses an identity design metric:

```
kern <- make_sdam_design_kernel()
round(kern$K, 3)
```

```
               recog_low_mid recog_high_sem nback_low_mid nback_high_sem
recog_low_mid          0.250          0.083         0.083          0.000
recog_high_sem         0.083          0.250         0.000          0.083
nback_low_mid          0.083          0.000         0.250          0.083
nback_high_sem         0.000          0.083         0.083          0.250
```

Diagonal cells are similar to themselves; off-diagonal entries are zero, so latent directions are not smoothed or coupled by the factorial design kernel. This is the unconstrained comparison baseline for the structured DKGE report.

3 Component summary

Table 2: Variance carried by each DKGE component (q = 4, kept rank = 3).

component	sdev	variance	variance_percent	cumulative
1	638.7623	408017.24	71.00	71.00
2	302.6972	91625.59	15.94	86.94
3	273.9382	75042.13	13.06	100.00

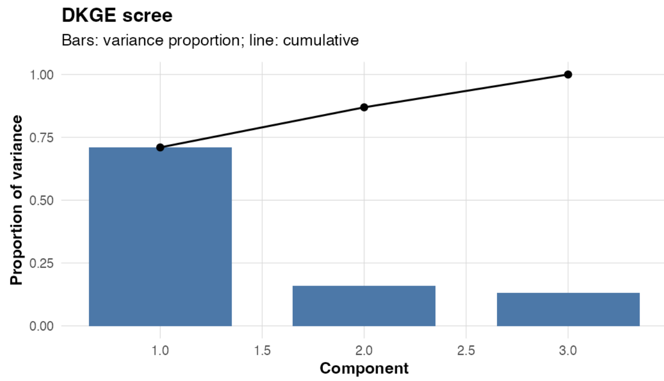

Figure 1: Scree plot of the DKGE compressed-covariance eigenvalues.

4 Design saliences

The cell-space loading matrix `K %*% U` is the DKGE analogue of PLS design saliences. Each panel below is one component; cells are laid out as `task x measure` so that main effects and interaction structure are visible at a glance.

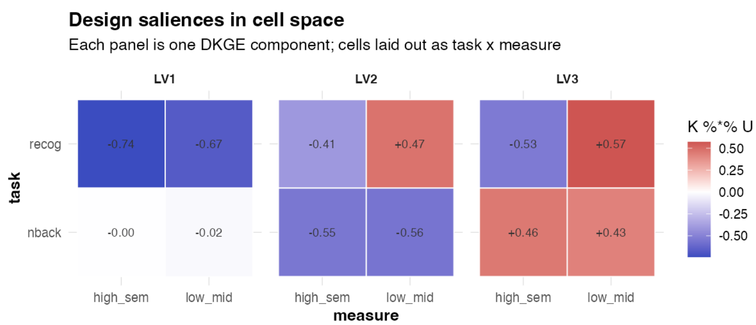

Figure 2: Cell-space loadings for each component.

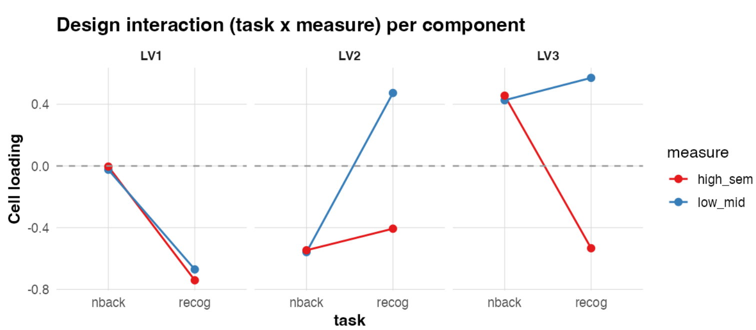

Figure 3: Interaction view: lines connect cells sharing a measure across tasks. Parallel lines indicate additive structure; crossing lines indicate task x measure interaction.

5 Subject-level diagnostics

DKGE's MFA-style subject weighting downweights noisy subjects automatically. Energy on each component shows where each subject's variance lands.

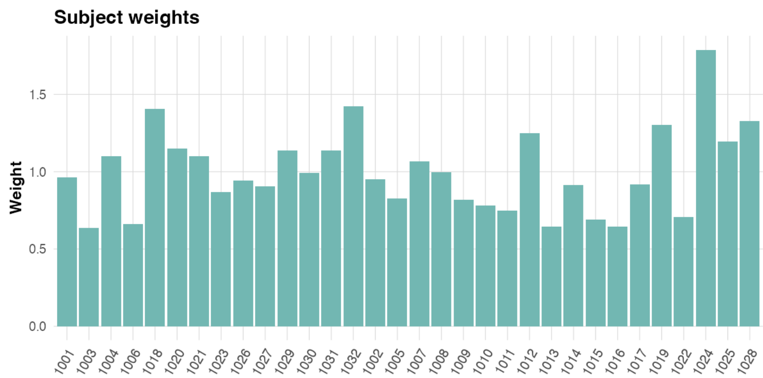

Figure 4: Subject weights and per-component participation energy.

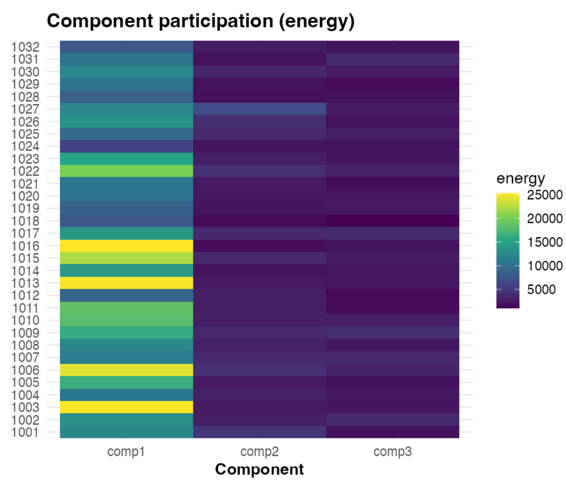

Figure 5: Subject weights and per-component participation energy.

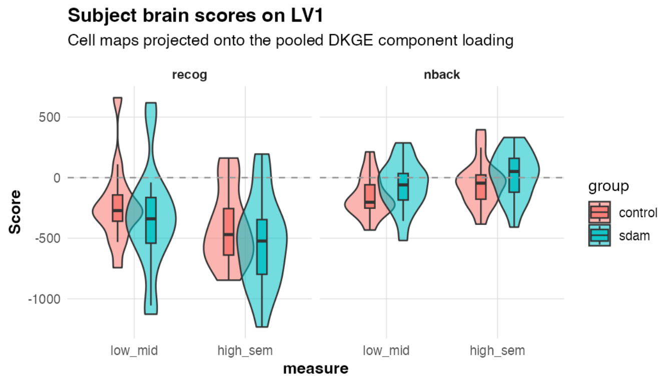

Figure 6: Subject 'brain scores' on LV1, by task (facets) x measure (x axis) and group (fill). Equivalent role to PLS scores plotted by condition.

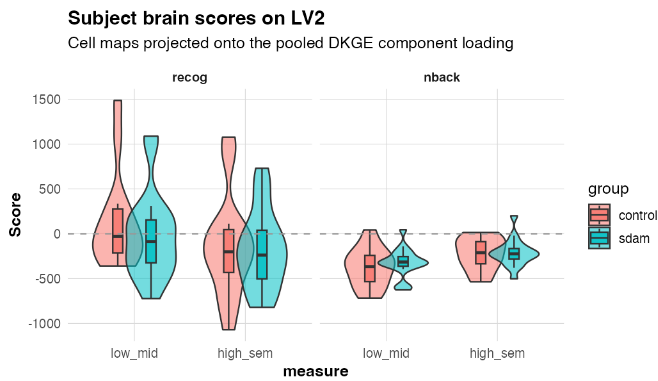

Figure 7: Subject brain scores on LV2.

The score plots project each subject-cell map onto a pooled DKGE component loading map, so they are the DKGE score-space analogue of PLS brain-score plots. The table below tests whether group differs on each component-level cell contrast.

Table 3: Group differences in DKGE component score-space contrasts. The task, measure, and task:measure rows are the group-by-within-subject effects for each component.

	component	contrast	mean_control	mean_sdam	diff_sdam_minus_control	statistic	p	p_adjusted
7	LV1	task	220.1000	409.6133	189.5133	-1.4939	0.1457	0.8740
4	LV1	measure	-31.7593	-54.0402	-22.2809	0.3745	0.7117	0.9120
10	LV1	task:measure	144.0125	140.9508	-3.0617	0.0510	0.9597	0.9597
1	LV1	grand_mean	-208.0912	-230.4550	-22.3637	0.3233	0.7487	0.9120
8	LV2	task	-296.0329	-171.0749	124.9580	-0.7659	0.4508	0.9120
5	LV2	measure	-62.2187	-25.6786	36.5401	-0.5336	0.5979	0.9120

	component	contrast	mean_control	mean_sdam	diff_sdam_minus_control	statistic	p	p_adjusted
11	LV2	task:measure	214.3974	137.0343	-77.3631	1.1000	0.2822	0.8919
2	LV2	grand_mean	-153.3482	-184.4016	-31.0534	0.3089	0.7600	0.9120
9	LV3	task	174.8588	184.6897	9.8309	-0.1479	0.8838	0.9597
6	LV3	measure	-30.0651	-56.9600	-26.8949	1.0615	0.2973	0.8919
12	LV3	task:measure	-1.0303	35.6325	36.6628	-1.5529	0.1319	0.8740
3	LV3	grand_mean	41.8939	56.1727	14.2788	-0.4147	0.6827	0.9120

6 LOSO contrast maps

Cross-fitted (leave-one-subject-out) contrast values give an unbiased per-subject voxel map for each design contrast. The summary statistics below are computed across the 32 subjects.

Table 4: Voxel statistics for each LOSO contrast: mean across subjects, z = mean / SE, and the number of voxels with |z| >= 3.

contrast	mean_min	mean_max	z_min	z_max	n_z_ge3
task	-0.026	0.046	-4.84	5.99	5144
measure	-0.008	0.015	-5.20	4.93	3803
task:measure	-0.016	0.009	-4.83	5.23	4022
grand_mean	-0.009	0.028	-3.54	6.11	5541

7 Bootstrap-ratio brain maps

For each contrast we resample the 32 LOSO subject vectors with replacement (200 draws) and report z = mean / SD at every voxel. This is the DKGE analogue of the PLS bootstrap-ratio map.

Table 5: Bootstrap z-statistic per voxel for each contrast.

contrast	min_z	max_z	pos_ge3	neg_le_neg3
task	-5.43	6.18	4007	1651
measure	-5.48	5.27	5273	178
task:measure	-5.36	5.42	182	5451

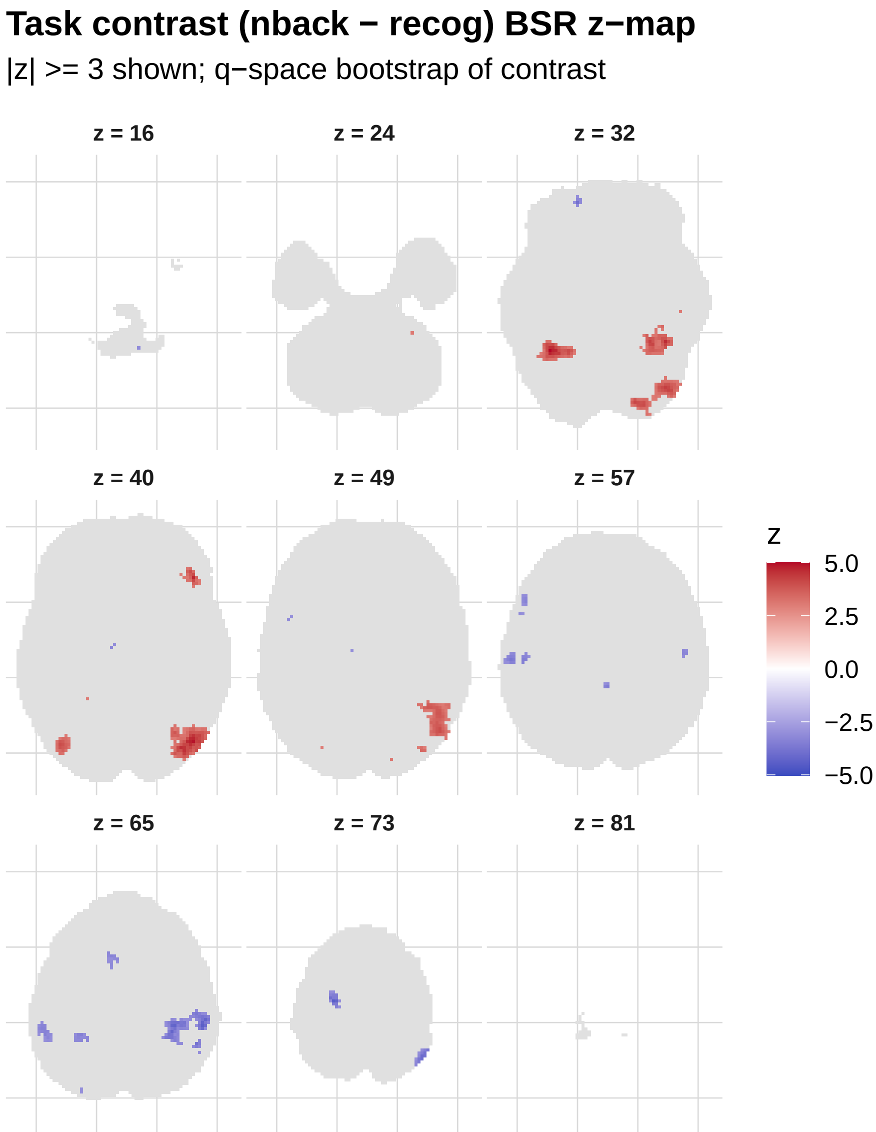

Figure 8: Task contrast (nback - recog) bootstrap z-map; nine evenly-spaced axial slices, |z| >= 3.

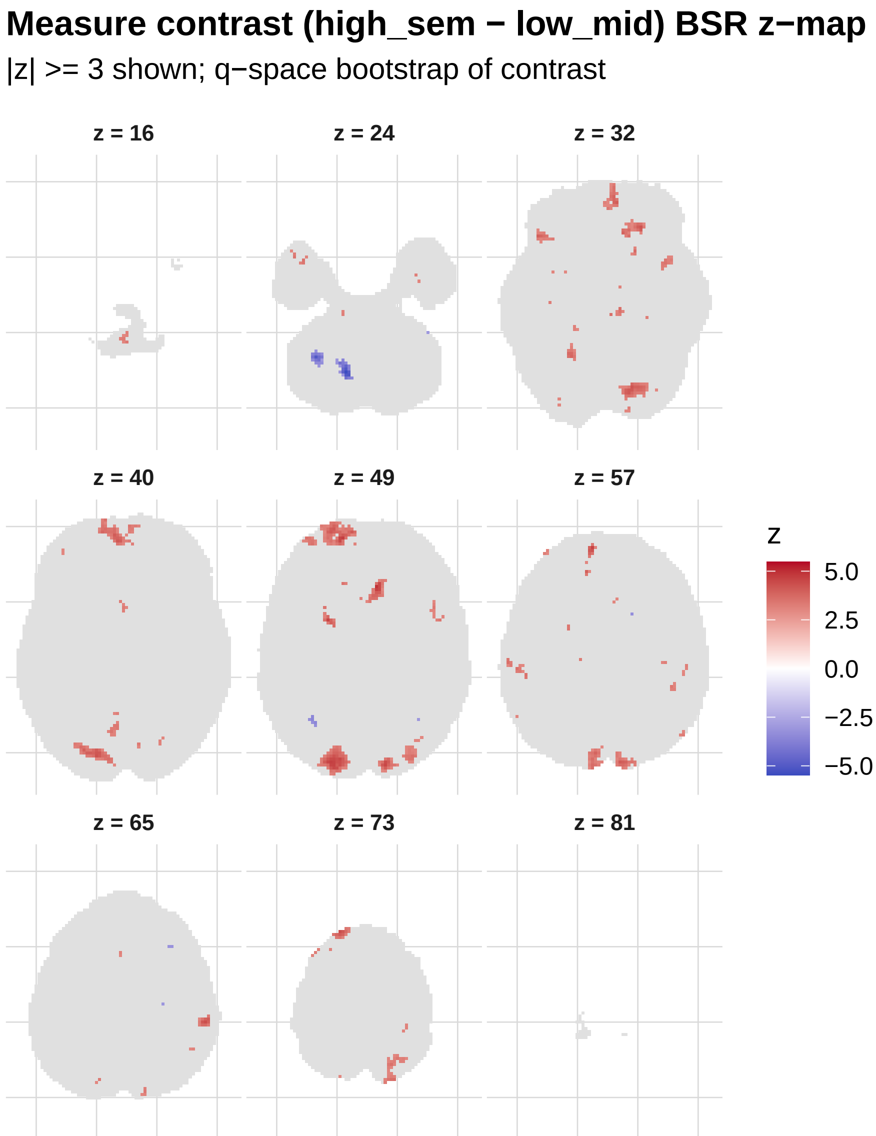

Figure 9: Measure contrast (high_sem - low_mid) bootstrap z-map; nine evenly-spaced axial slices, |z| >= 3.

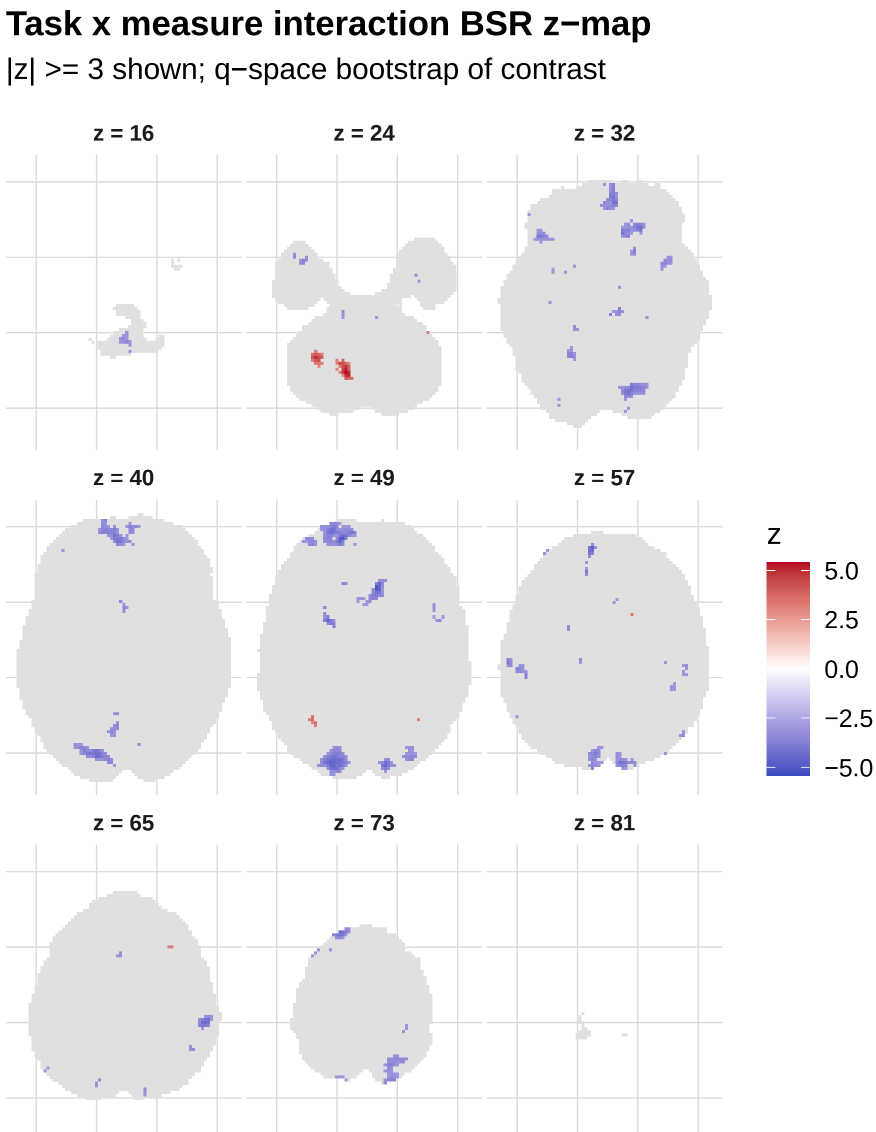

Figure 10: Task x measure interaction bootstrap z-map; nine evenly-spaced axial slices, |z| >= 3.

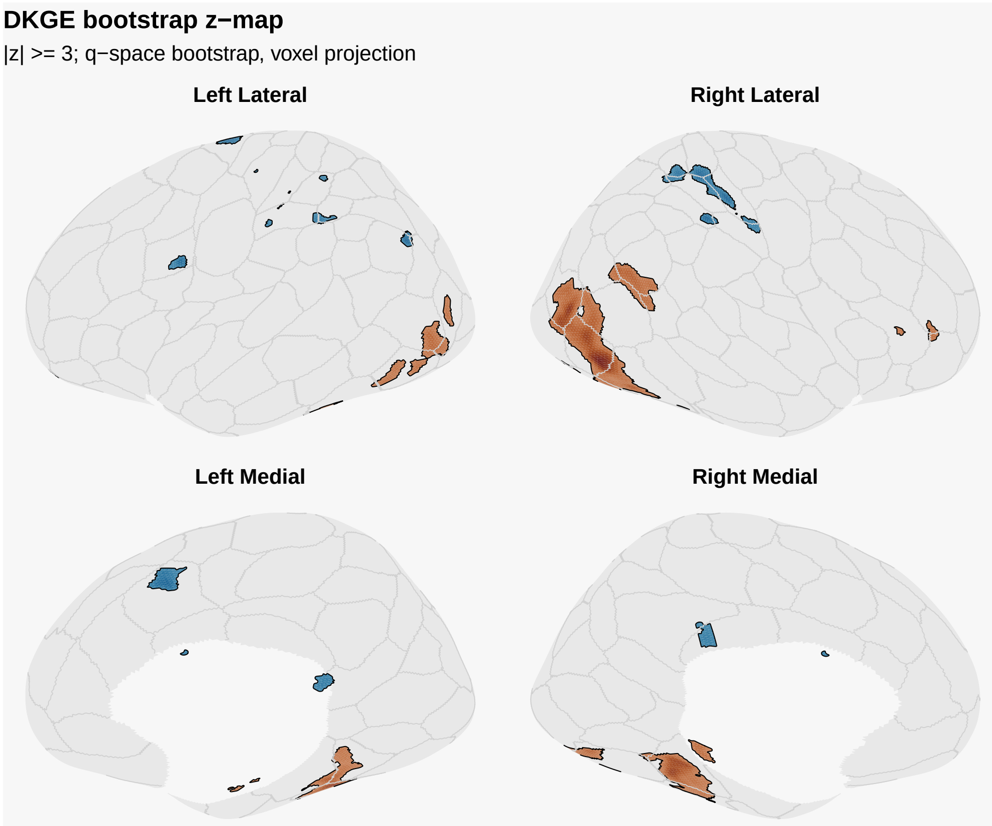

Figure 11: Task contrast bootstrap z-map projected onto a Schaefer 200x7 surface (fsaverage6).

8 Primary between-subject DKGE: control vs SDAM

The voxel-level contrast-map target Y is regressed on `~ group` via reduced-rank regression. Rows are subjects; columns are feature blocks formed by each cross-fitted repeated-measures contrast (`grand_mean`, `task`, `measure`, `task:measure`) crossed with voxels. This tests whether control and SDAM subjects differ in the spatial expression of the within-subject effects, rather than reducing each subject to three component means first.

This is the primary DKGE inference path for the PLS-like crossover question: fit a shared $q = 4$ task x measure DKGE model, build cross-fitted subject-level contrast targets, and test `group` as a between-subject predictor.

Table 6: Between-subject target dimensions: 32 subjects x 873,656 contrast-voxel features.

contrast	features
task	218414
measure	218414
task:measure	218414
grand_mean	218414

Table 7: Permutation test for the group term in the voxel-level contrast-map RRR.

	term	statistic	p	p_adjusted
group	group	3.9706	0.56	0.56

Table 8: Focused primary test: group effect on the cross-fitted task:measure contrast target only.

	term	statistic	p	p_adjusted	target_features
group	group	0.1656	0.842	0.842	218414

Table 9: Top featurewise group effects by maxT-adjusted p-value.

	contrast	feature	statistic	p	p_adjusted
54308	task	task:54308	0.0086	0.600	0.600
54309	task	task:54309	0.0084	0.596	0.611
54365	task	task:54365	0.0082	0.591	0.619
54364	task	task:54364	0.0082	0.596	0.623
49582	task	task:49582	0.0080	0.597	0.631
54255	task	task:54255	0.0079	0.597	0.634
59169	task	task:59169	0.0079	0.597	0.636
36431	task	task:36431	0.0079	0.584	0.637
63831	task	task:63831	0.0079	0.592	0.637
59224	task	task:59224	0.0079	0.598	0.637

9 q = 8 group-cell mean DKGE bridge to PLS

This is the q = 8 cell-mean analysis in which `group` enters the analysis up front. It is not the strict partial-effect subject-level fit. The target is an 8-row matrix of observed group-cell means: for each group, task, and measure row, subject maps are averaged within group before the decomposition.

Mean-centered task PLS decomposes this kind of group/condition cell-mean object, so this is the DKGE bridge that is closest to the PLS LV display. We build the 8-row `group x task x measure` target through `dkge_aggregate_target()`, apply a DKGE design kernel over those aggregate rows, and decompose the resulting cell-mean object. It is a secondary PLS-facing analysis; the primary DKGE inference remains the q = 4 subject-level model above.

Permutation and bootstrap inference still resample subjects, not the eight aggregate rows. For every draw, the subject labels or subject sample are changed, the group x task x measure means are recomputed, the aggregate decomposition is refit, components are aligned back to the observed fit, and the group:task:measure statistic is reevaluated. The bridge statistic below is a selected-component, singular-value-scaled salience contrast on the cell-mean object. It is useful for comparing the shape of DKGE and PLS cell-mean decompositions, but it is not a replacement for the primary subject-level `dkge_between_*` test above.

Table 10: q = 8 group-cell mean DKGE bridge target and fit settings.

rows	features	aggregate	inference_unit	center	rank	tested_component
8	218414	mean	subject	column	3	LV2

Table 11: The eight observed group x task x measure cell means decomposed by the bridge analysis.

group	task	measure
control	nback	high_sem
control	nback	low_mid
control	recog	high_sem
control	recog	low_mid
sdam	nback	high_sem
sdam	nback	low_mid
sdam	recog	high_sem
sdam	recog	low_mid

Table 12: Singular values of the q = 8 group-cell mean DKGE bridge.

component	singular_value
LV1	5.3944
LV2	4.6092
LV3	3.2044

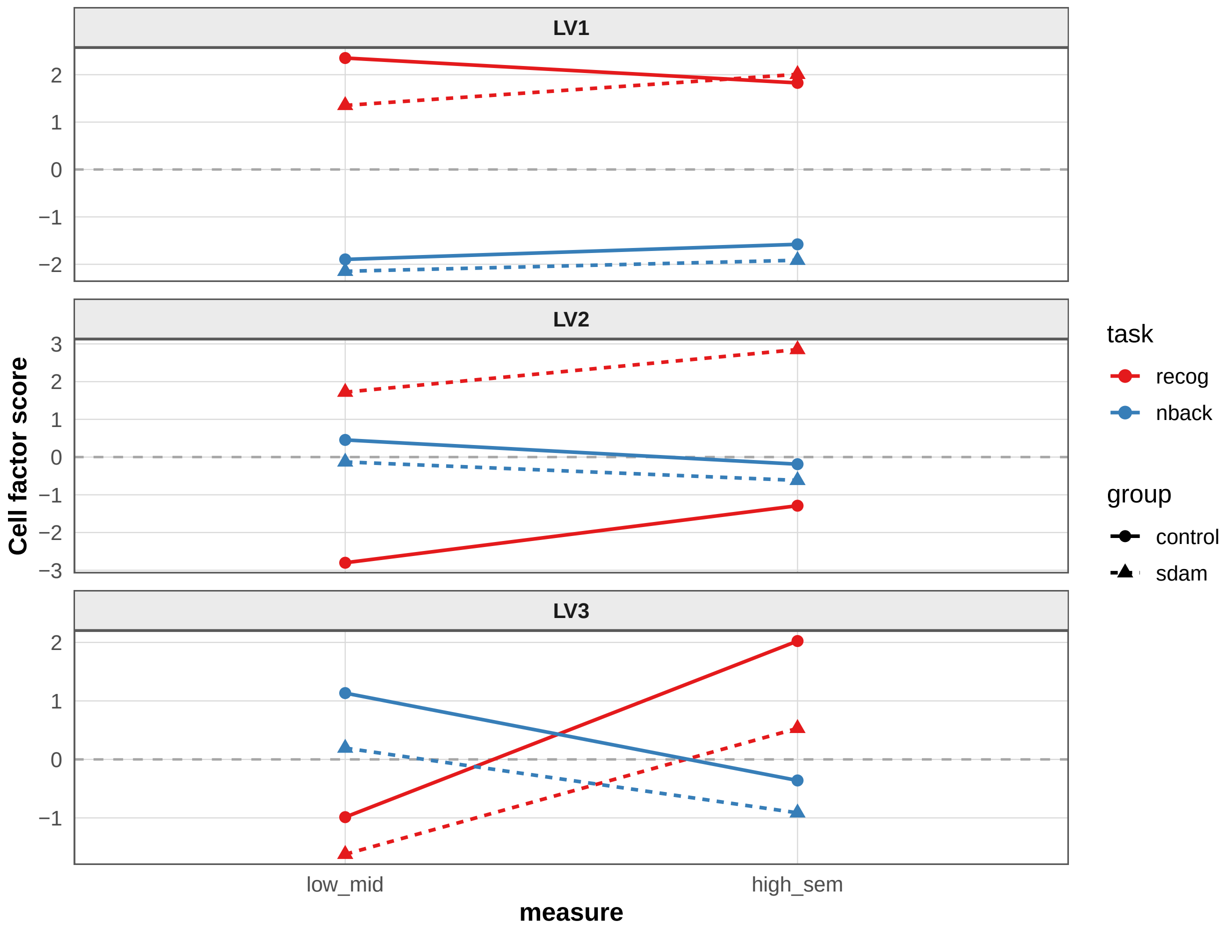

Figure 12: q = 8 group-cell mean DKGE bridge factor scores. Each LV has its own boxed panel; task is color and group is line type/point shape.

Table 13: LV-wise q = 8 bridge cell-score contrasts from subject bootstrap. Intervals excluding zero mark visually large effects that were stable under subject resampling.

contrast	component	observed	bootstrap_low	bootstrap_high	excludes_zero
task	LV1	-3.7693	-5.8532	4.5953	FALSE
task	LV2	-0.2402	-4.5462	4.4820	FALSE
task	LV3	0.0271	-1.5580	1.6710	FALSE
measure	LV1	0.1705	-1.4062	0.9307	FALSE
measure	LV2	0.3775	-1.0258	0.8455	FALSE
measure	LV3	0.6408	0.4174	1.5535	TRUE
task:measure	LV1	0.1048	-1.5067	2.3462	FALSE
task:measure	LV2	-0.9434	-1.3906	1.7871	FALSE
task:measure	LV3	-1.9409	-2.5523	-1.0741	TRUE
group	LV1	-0.3513	-4.2567	4.0017	FALSE
group	LV2	1.9128	-2.4073	3.3623	FALSE
group	LV3	-0.9052	-2.1385	0.7234	FALSE
group:task	LV1	0.0582	-4.1288	4.5796	FALSE

contrast	component	observed	bootstrap_low	bootstrap_high	excludes_zero
group:task	LV2	-2.4191	-3.6196	2.4772	FALSE
group:task	LV3	0.1586	-1.4958	0.8230	FALSE
group:measure	LV1	0.2735	-0.6131	1.0896	FALSE
group:measure	LV2	-0.0573	-0.6278	0.8183	FALSE
group:measure	LV3	-0.1178	-1.2595	0.9691	FALSE
group:task:measure	LV1	-0.3164	-1.1881	0.5889	FALSE
group:task:measure	LV2	0.1341	-0.8256	0.5530	FALSE
group:task:measure	LV3	0.3114	-0.7597	1.3204	FALSE

Table 14: Subject-resampling inference for the $q = 8$ group-cell mean DKGE bridge group:task:measure statistic. The statistic is the selected component's singular-value-scaled salience contrast.

tested_component	statistic	permutation_p	permutation_B	bootstrap_low	bootstrap_high	bootstrap_B	near_tie_draws	weak_alignment_draws	min_alignment_cosine
LV2	0.1341	0.585	999	0.0096	0.9236	200	0	16	0.3409

Alignment diagnostic: 16 permutation draws had component alignment cosine below 0.50 (minimum 0.341). LV-specific bridge inference should therefore be treated as descriptive unless the statistic is changed to an omnibus/subspace target.

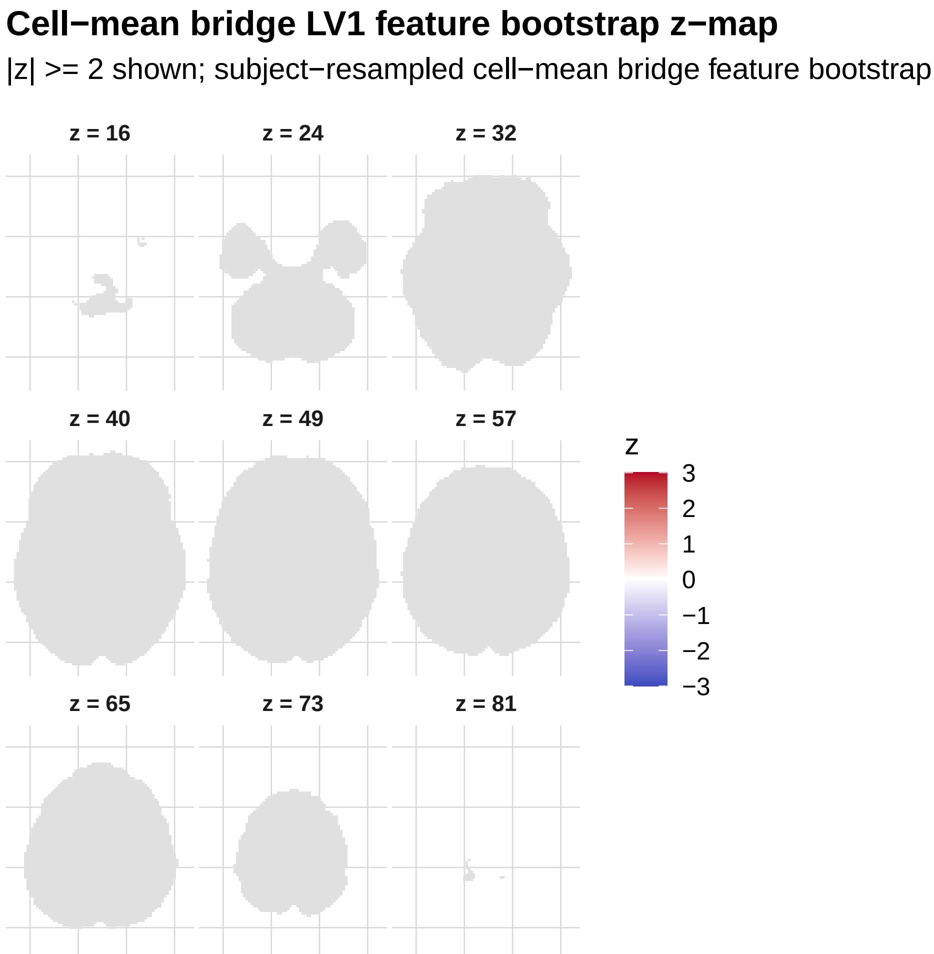

Figure 13: Bootstrap-ratio feature maps for the q = 8 group-cell mean DKGE bridge LVs. Maps are built by resampling subjects, recomputing group-cell means, refitting, aligning components, and accumulating feature-space component scores.

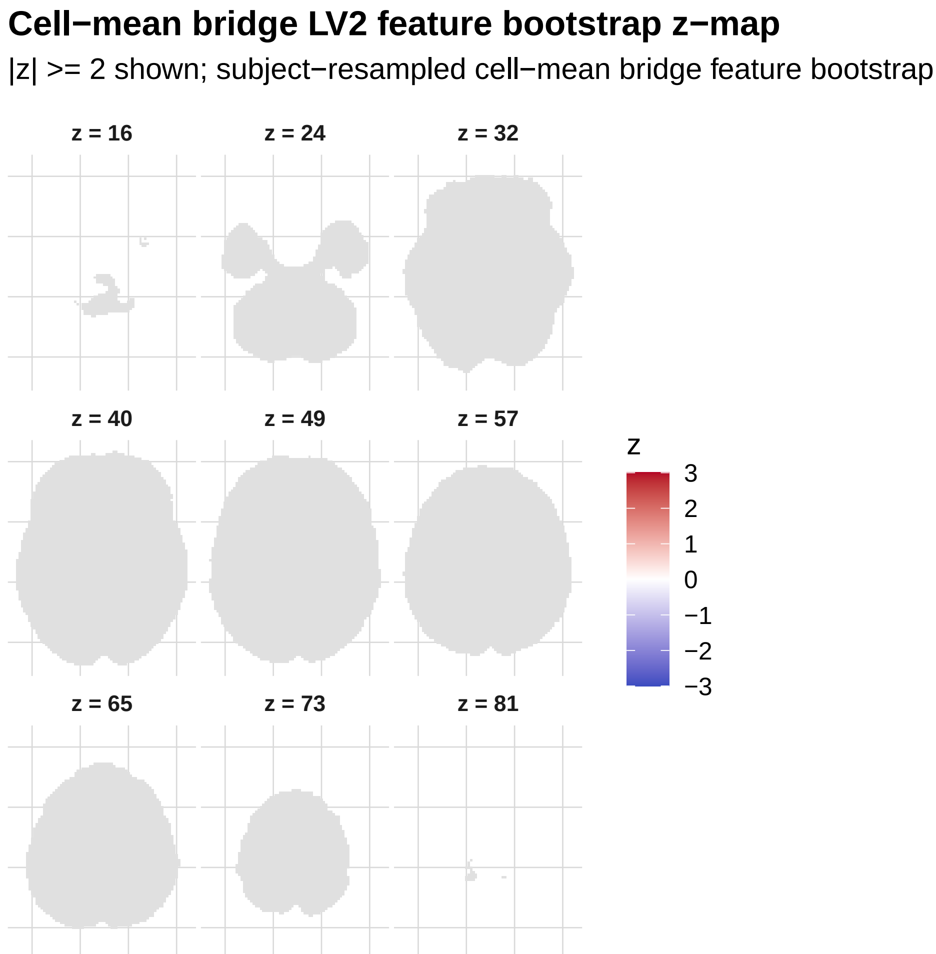

Figure 14: Bootstrap-ratio feature maps for the q = 8 group-cell mean DKGE bridge LVs. Maps are built by resampling subjects, recomputing group-cell means, refitting, aligning components, and accumulating feature-space component scores.

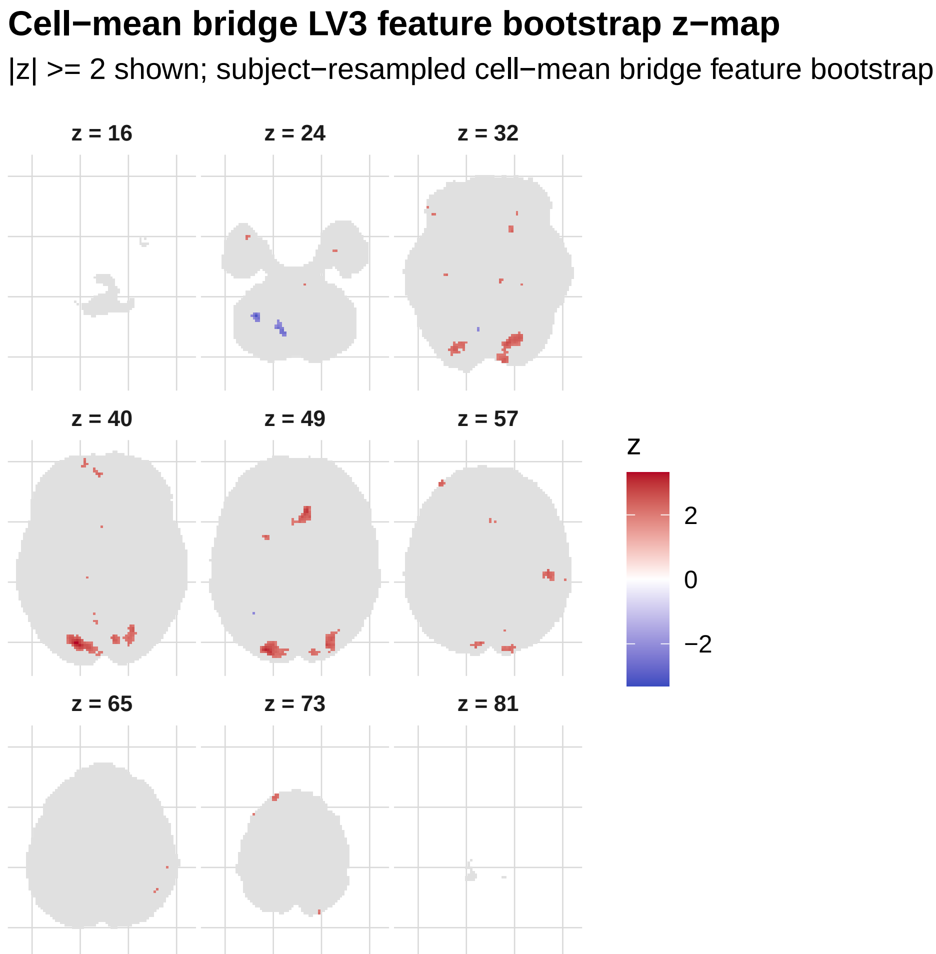

Figure 15: Bootstrap-ratio feature maps for the q = 8 group-cell mean DKGE bridge LVs. Maps are built by resampling subjects, recomputing group-cell means, refitting, aligning components, and accumulating feature-space component scores.

10 Strict block-diagonal q = 8 partial-effect fit

The strict block-diagonal q = 8 partial-effect fit is intentionally omitted from the main figures. In that model each subject observes only the four rows for their own group, and the block-diagonal group kernel leaves the control and SDAM row spaces disconnected. Its components are therefore group-local axes sorted by eigenvalue, not shared group x task x measure latent variables. Those component plots are not interpretable as a PLS LV2 analogue, so they are not shown here.

11 Stratified basis diagnostic

The voxel-level group test above assumes the pooled cell-space basis is a reasonable shared design space. As a diagnostic, DKGE is also fit separately in controls and SDAM, then compared with

K-metric principal angles. These angles are descriptive; a stronger claim would compare them against within-group split-half or bootstrap stability.

A geometric note on rank choice. Two rank-`r` subspaces of a `q`-dimensional ambient space share at least `max(0, 2r - q)` dimensions by the dimension formula, so the first that-many principal angles are forced to zero regardless of the data. With `q = 4` and pooled rank 3, two of the three angles would be geometric necessities. The diagnostic therefore caps the *stratified* rank at `floor(q / 2) = 2` so every reported angle is informative; the pooled fit still uses the full requested rank.

Table 15: Principal angles between control-only and SDAM-only DKGE bases (q = 4, stratified rank = 2, forced overlap = 0).

component	group_a	group_b	angle_rad	angle_deg
LV1	control	sdam	0.0016	0.09
LV2	control	sdam	0.3031	17.37

12 Voxel-level group difference (sdam - control) on LOSO contrasts

Table 16: Per-voxel two-sample z statistic for the SDAM minus control difference of each LOSO contrast.

contrast	n_sdam	n_control	mean_diff_min	mean_diff_max	z_min	z_max	n_z_ge3
task	18	14	-0.046	0.040	-4.92	4.23	576
measure	18	14	-0.023	0.018	-4.41	5.51	878
task:measure	18	14	-0.019	0.023	-5.65	4.48	869
grand_mean	18	14	-0.020	0.022	-5.10	3.97	286

13 Reproducibility

```
R version: R version 4.5.1 (2025-06-13)
DKGE seed: 20260509
Mask voxels: 218414
Subjects (control / sdam): 14 / 18
```